EXPERIMENT No 3: Creating a Traffic Light with Mixly

Student Guide

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Objective:

In this experiment, you will learn how to simulate a basic traffic light system using Mixly and an Arduino. You will control three LEDs (representing red, yellow, and green traffic lights) and program them to follow a standard traffic light sequence.

Materials Needed:

- Arduino Uno
- 3 LEDs (Red, Yellow, Green)
- 3 Resistors (220Ω or 330Ω)
- Breadboard
- Jumper wires
- USB Cable (for connecting Arduino to your computer)
- Computer with Mixly software installed

What is a Traffic Light System?

A traffic light system is used to control vehicle and pedestrian traffic at intersections. It typically cycles between red, yellow, and green lights to signal when cars should stop, slow down, or go.

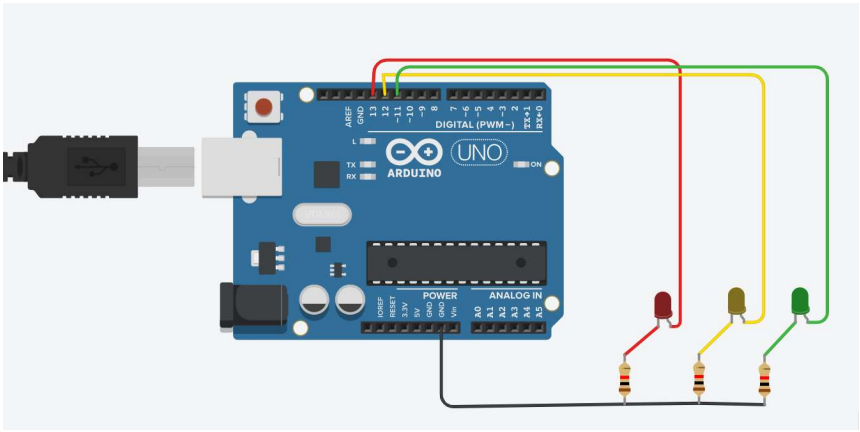
Steps to Create a Traffic Light System Using Mixly

Step 1: Set Up the Circuit

1. Connect the LEDs to the Breadboard:
   - Place the Red, Yellow, and Green LEDs on the breadboard. Make sure to note that:
     - Red LED (pin 13) will represent "Stop."
     - Yellow LED (pin 12) will represent "Caution" (prepare to stop).
     - Green LED (pin 11) will represent "Go."
2. **Add Resistors:**
   - Connect a 220Ω resistor in series with each LED's longer leg (anode) to limit the current and protect the LEDs.

3. **Connect the LEDs to the Arduino:**
   - Connect the long leg (anode) of each LED to the appropriate digital pins:
     - Red LED → pin 13
     - Yellow LED → pin 12
     - Green LED → pin 11

- Connect the short leg (cathode) of each LED to the GND (ground) pin of the Arduino.



Step 2: Open Mixly and Set Up the Program

1. **Open Mixly:**
   - Launch the Mixly program on your computer.
   - Select the correct board (Arduino Uno) and port in Mixly's settings.
2. **Create the Traffic Light Sequence Using Blocks:**

Red Light (Stop):

- Drag a "Control" block to set digital pin 13 to HIGH (this will turn the red LED on).
- Add a "Time" block with a delay of 5000 milliseconds (5 seconds).

Yellow Light (Caution):

- Add a "Control" block to set digital pin 12 to HIGH (this will turn the yellow LED on) and set digital pin 13 to LOW (turn off red LED).
- Add a "Time" block with a delay of 2000 milliseconds (2 seconds).

Green Light (Go):

- Add a "Control" block to set digital pin 11 to HIGH (this will turn the green LED on) and set digital pin 12 to LOW (turn off yellow LED).
- Add a "Time" block with a delay of 5000 milliseconds (5 seconds).

Return to Red Light:

- Set digital pin 11 to LOW (turn off green LED) and repeat the sequence by setting digital pin 13 to HIGH for the red light.

3. **Loop the Program:**
   - Add a "Forever" loop from the Control category to ensure the traffic light sequence runs continuously.
   - Place all the block commands inside the "Forever" loop.

Step 3: Upload the Program to the Arduino

1. **Check Your Connections:**
   - Ensure that all the LEDs are connected correctly to the breadboard and Arduino.
2. **Upload the Code:**
   - Once your block-based program is complete, click the Upload button in Mixly.
   - Wait for the program to compile and upload to the Arduino.

Step 4: Observe the Traffic Light Simulation

- After uploading, the red, yellow, and green LEDs should cycle through the traffic light sequence:
  - Red on for 5 seconds (Stop).
  - Yellow on for 2 seconds (Caution).

- Green on for 5 seconds (Go).
- Then the cycle repeats continuously.

Explanation of the Program:

- Pin 13 (Red): Lights up to signal "Stop" for 5 seconds.
- Pin 12 (Yellow): Lights up to signal "Caution" for 2 seconds.
- Pin 11 (Green): Lights up to signal "Go" for 5 seconds.
- The delays ensure that each light stays on for the appropriate amount of time before switching to the next.

Conclusion:

In this experiment, you successfully created a simple traffic light system using Mixly and an Arduino. This hands-on activity helps you understand basic programming and electronic components, as well as how to simulate real-world systems using LEDs.

Challenge:

- Try adding more LEDs to create a pedestrian signal alongside the traffic light.
- Modify the timing of the lights to make the system more complex.

This activity teaches you about controlling multiple LEDs with Mixly and simulating systems like a traffic light.